CS2600

THE UNIVERSITY OF WARWICK
Summer Examinations 2017/18
Algorithms

Time allowed: 2 hours.

Answer **THREE** questions: **ONE** from Section A and **TWO** from Section B.

Make it clear on the cover of the answerbook which three questions you have selected.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on **each** answer book.

Calculators are not required and not permitted.

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continued

Section A

Answer **ONE** question from this section.

1. (a) **[6 points]** Use the Karatsuba integer multiplication algorithm to multiply the binary integers 1101 and 1010. (Your answer should show how the algorithm works given this input.)

   (b) **[6 points]** Give the definition of a polynomial-time reduction.

   (c) **[8 points]** Suppose the greedy algorithm is run on some instance of the set cover problem with universe (ground set) $\{e_1, \ldots, e_n\}$. Assume that on each step the number of elements that are not yet covered shrinks at least by a factor of 0.9. Prove that the cover produced by the algorithm has size $O(\log n)$.

2. An undirected graph $G = (V, E)$ *has the half-clique property* if there exists a subset $S \subseteq V$ with $|S| \geq |V|/2$ such that every two distinct vertices $u, v \in S$ are connected by an edge in E. The HALFCLIQUE problem is to decide, given G, whether G has the half-clique property.

   (a) **[3 points]** Draw a graph with 7 vertices and 7 edges that has the half-clique property.

   (b) **[5 points]** Draw a graph with 7 vertices and 15 edges that does not have the half-clique property.

   (c) **[12 points]** Prove that HALFCLIQUE is NP-complete.

continued

Section B

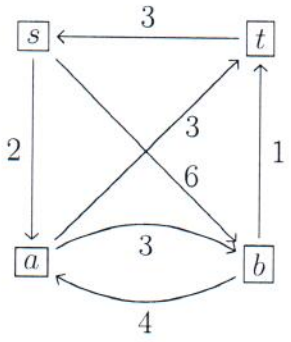
Answer **TWO** questions from this section.

continued

3. (a) **[6 points]** Let (G, s, t, c) be a flow network where $G = (V, E)$ is a directed graph, s and t are two distinct vertices of G called source and sink, and the function $c\colon E \to \mathbb{R}_{>0}$ specifies the capacities of the edges.

   Give the definitions of an (s,t)-flow and an (s,t)-cut in the network.

   (b) **[6 points]** For the following network, write a linear program encoding the maximum flow problem. Each edge is labelled by a positive integer which denotes the capacity of the edge. Vertices s and t are the source and the sink, respectively.



   (c) **[6 points]** What is the value of a maximum (s,t)-flow in the network above?

   (d) **[8 points]** Suppose the standard Union–Find data structure is used to maintain a collection of disjoint sets. Assume that union by rank with path compression is used.

   Prove that, whenever a `find(·)` operation traverses an edge (link) (u, v), it is the case that the rank of u is strictly lower than the rank of v.

   You can assume without proof that the rank of any node becomes fixed when the node stops being a root.

A software repository has n packages, numbered 1 through n. A dependency list is a set D of pairs (i, j) such that package j cannot be installed without installing package i.

   (e) **[6 points]** Suppose $n = 8$ and D is the set of the following pairs:

$$(1,2)\quad (3,2)\quad (3,4)\quad (4,1)\quad (5,1)\quad (5,4)\quad (6,8)\quad (8,3)$$

   In this example, in order to install package 1, it is necessary to install six packages in total (including package 1 itself). Which packages are these?

   (f) **[8 points]** Give a linear-time algorithm that takes as input a number n and a dependency list D, and produces the list of all packages that are necessary to install package 1.

   Justify the correctness of your algorithm and prove that it runs in linear time.

continued

4. (a) **[7 points]** Suppose the radix sort algorithm is run on the following array of 3-digit positive decimal integers:

787 905 681 99 672 174 557 441 19

Draw a table showing the contents of the array after each iteration of the main loop of the radix sort.

(b) **[5 points]** For each of the following statements, indicate whether it is true or false. In all five statements, n is the number of elements in the input array.

i. Quicksort has worst-case running time $O(n \log n)$ if the median element is chosen as pivot in each recursive invocation.

ii. Quicksort has worst-case running time $O(n^2)$ if the median element is chosen as pivot in each recursive invocation.

iii. Quicksort has worst-case running time $\Theta(n^2)$ if the first element is chosen as pivot in each recursive invocation.

iv. For every input, Quicksort has expected running time $\Theta(n^2)$ if the pivot is chosen uniformly at random (from the array) in each recursive invocation.

v. Every algorithm for sorting integers from the set $\{1, 2, \ldots, n^{10}\}$ has worst-case running time $\Omega(n \log n)$.

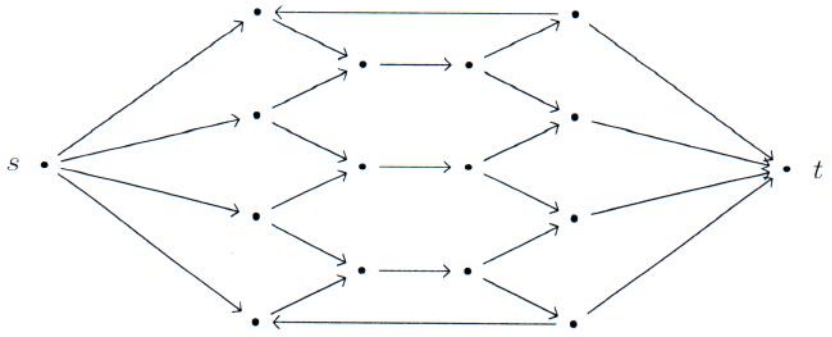
(c) **[8 points]** Suppose the Mergesort procedure is modified as follows. If the array has at most 100 elements, *insertion sort* is run instead, otherwise Mergesort proceeds as usual. This modification applies to each recursive invocation.

Analyze the worst-case running time of the resulting algorithm. Assume that insertion sort runs in time $\Theta(n^2)$ where n is the number of elements in the input.

Two paths in a directed graph are *edge-disjoint* if they do not share any edges.

(d) **[5 points]** Find 3 pairwise edge-disjoint paths from s to t in the graph below.

(e) **[7 points]** Prove that the graph below does not have 4 pairwise edge-disjoint paths from s to t.



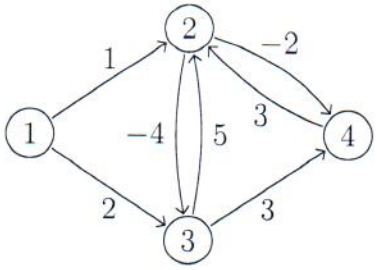
(f) **[8 points]** Given a linear-time algorithm that, given a directed graph $G = (V, E)$ in the adjacency lists representation and two distinct vertices $s, t \in V$, determines whether G has 3 pairwise edge-disjoint paths from s to t.

Justify the correctness of your algorithm and prove that it runs in linear time.

continued

5. In this question, you will develop *Johnson's algorithm* for the all-pairs shortest paths problem.

   (a) **[7 points]** In the all-pairs shortest paths problem, the input is a directed graph $G = (V, E)$, $V = \{1, \ldots, n\}$, with edge weights $\ell\colon E \to \mathbb{R}$. The output is a matrix $D = (d_{st})$ of size $n \times n$ where d_{st} denotes the length of some shortest path from s to t in G. Solve the all-pairs shortest path problem for the following graph:



   In all sub-questions below, assume that G has no negative cycles.
   Also assume (for simplicity) that all vertices are reachable from vertex 1.

   (b) **[5 points]** Johnson's algorithm uses as subroutines Dijkstra's algorithm and the Bellman—Ford algorithm for shortest paths.

   For each of these two algorithms, specify (i) what is given as input, (ii) what is the required output, and (iii) what is the worst-case running time.

   Note that you can solve the all-pairs shortest paths problem by running the Bellman—Ford algorithm n times. What is the running time for this approach?

   (c) **[9 points]** A *potential function* is a mapping $f\colon V \to \mathbb{R}$. Given a potential function f, define new edge weights ℓ' by the following rule: $\ell'(u, v) = \ell(u, v) + f(u) - f(v)$. Let G' denote the same graph (V, E) with edge weights ℓ' instead of ℓ.

   Show that for every path π from s to t it holds that $\ell'(\pi) = \ell(\pi) + f(s) - f(t)$ where $\ell(\pi)$ and $\ell'(\pi)$ denote the length of π in G and in G', respectively.

   Prove that, for every pair $s, t \in V$, the set of shortest paths from s to t is the same in G' as in G.

   A potential function is *convenient* if all edge weights in G' are nonnegative, that is, if $\ell'(u, v) \geq 0$ for all $(u, v) \in E$.

   (d) **[8 points]** Give an algorithm that, given G, ℓ, and a convenient potential function, computes the distances d_{st} for all pairs $(s, t) \in V \times V$. Your algorithm should run in time $O(n \cdot (n + m) \cdot \log n)$ where $m = |E|$.

   (e) **[11 points]** For an arbitrary G and ℓ, find a convenient potential function f and prove that it is convenient. Describe an algorithm that, given G and ℓ, computes f in time $O(n \cdot m)$ where $m = |E|$.

   Johnson's algorithm is a combination of the algorithms for (e) and (d).

(end)